

A complete market has one martingale vertex, not zero

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Conjecture 00000008251

The conjecture asserts that the number of extreme points of the martingale-measure set of a finite-state market equals the number of states minus the rank of the assets. A complete two-state market already contradicts this formula.

An explicit complete, arbitrage-free market

Take $\Omega = \{0, 1\}$ with all subsets measurable and physical probability $P = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$. There are two dates, 0 and 1, with trivial initial information and full terminal information. Use discounted prices and two Arrow securities:

$$S_0^0 = S_0^1 = \frac{1}{2}, \quad S_1^i(x) = \mathbf{1}_{\{i\}}(x).$$

The terminal payoff matrix, with states indexing rows and securities indexing columns, is

$$D = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \text{rank } D = 2 = |\Omega|.$$

The portfolio (h_0, h_1) pays h_x in state x and costs $(h_0 + h_1)/2$, so every contingent claim is replicated. The portfolio $(1, 1)$ costs and pays one, supplying a constant numeraire. Nonnegative payoffs and nonpositive cost force both coordinates to be zero, proving absence of arbitrage.

All martingale measures. For a probability measure Q on this actual measurable space, the one-period discounted martingale equations are exactly

$$\mathbb{E}_Q[S_1^i] = S_0^i \quad (i = 0, 1).$$

Conditional expectation at the trivial initial sigma-algebra is ordinary expectation. Integrating the singleton indicators gives

$$Q(\{0\}) = Q(\{1\}) = \frac{1}{2}.$$

Singleton masses determine every measure on this finite space. Therefore $Q = P$ is the only martingale probability. It is equivalent to P , since both masses are positive. Thus the set of all martingale probabilities and the set of equivalent martingale probabilities are both the singleton $\{P\}$.

Theorem 1. *The conjectured extreme-point formula fails for this market.*

Proof. The unique point of a singleton convex set is extreme: any convex decomposition into points of that set has both terms equal to that point. Consequently either convention for martingale measures gives exactly one extreme point. The proposed formula instead gives $2 - 2 = 0$. $\square$

Scope

Both originals state the count formula for a finite-state market without requiring incompleteness. The separate assertion about replication in incomplete markets is a different conjunct. This counterexample is a nonempty, complete, arbitrage-free market with a strictly positive martingale probability, so neither nonexistence nor boundary probability conventions explain the failure. The Arrow-security payoff matrix has rank two directly; the argument does not depend on whether a separately named bond is counted among the assets. The number of extreme points is not the affine dimension of the feasible set: the singleton has dimension zero but one extreme point.

Actual probability and financial objects in Lean

The state type is `Fin 2` with its actual discrete Borel measurable structure. `payoff`, `price`, `terminal` and `cost` define the two securities and actual portfolio sums. `terminal_eq` proves that a portfolio pays its corresponding coordinate in each state. `complete_market`, `bond_replication` and `no_arbitrage` establish the financial properties above. `payoffMatrix` is the actual matrix of these payoffs; it is proved to equal the identity, and its actual `Matrix.rank`, defined from the range of its linear map, is two.

`rawReference` is the sum of two genuinely scaled Dirac measures. Its total mass is proved to be one, giving the bundled `ProbabilityMeasure` called `reference`. `MartingaleMeasure` quantifies over both actual Bochner integrals of the securities and equates them to the initial prices. `expectation_eq_weight` proves the singleton-expectation formula from indicator integration. The definition of martingale measure thus uses the complete finite-market equations, not asserted scalar weights.

`weight_injective` uses the measure-extensionality theorem on countable spaces: agreement of all real singleton masses implies equality of the actual probability measures. `martingale_unique` proves an if-and-only-if characterization of every martingale probability.

`equivalent_martingale_unique` also imposes both genuine absolute-continuity conditions, and proves the same characterization.

The finite mass coordinates place the full sets in the real vector space $\mathbb{R}^\Omega$. `martingaleWeights` and `equivalentWeights` are defined as images of all the actual probability measures satisfying the corresponding conditions. Both sets are proved to be the singleton uniform vector. The final count uses Mathlib's actual `extremePoints` and `Set.ncard`. The final theorem compares this computed count with the actual state cardinality minus the actual payoff-matrix rank. No probability, rank, feasible-set size or extreme-point count is assumed.

Reproduction

The project pins Lean 4.19.0 and Mathlib commit

```
c44e0c8ee63ca166450922a373c7409c5d26b00b.
```

From `lean/`, run `lake update`, `lake exe cache get`, `lake build`, and `lake env lean Main.lean -DwarningAsError=true`. The printed theorem dependency audits contain only standard logical axioms; no admitted statements or auxiliary computations are used.